

Program: Electronics and Computer Engineering	
Course Code: 6501C	Course Title: Fundamentals of Photovoltaics and solar cell technology
Semester: 6	Credits: 4
Course Category: Open elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- The course aims to provide a basic understanding about photovoltaics and solar cell technology.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Semiconductor theory	2041	Basic Electronics	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Explain different renewable and non-renewable energy sources and its types	12	Understanding
CO2	Describe the fundamentals of solar cells.	12	Understanding
CO3	Explain solar cell characteristics	16	Understanding
CO4	Classify various photovoltaic cell technologies and describe the fabrication and formation of PV modules	18	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3	3					
CO3	3	3					
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain different renewable and non-renewable energy sources and its types		
M1.01	Explain the need of switching to Renewable Energy sources	2	Understanding
M1.02	Differentiate between Renewable & Non-renewable sources.	2	Understanding
M1.03	Describe the main sources of Renewable	4	Understanding
M1.04	Explain the Applications, Advantages & Disadvantages of Renewable Energy.	4	Understanding
Contents: Fundamentals of Sustainable Energy & Development Introduction to RE – Need of switching to Renewable Energy sources, Difference between Renewable & Non-renewable sources, Main sources – solar, wind, tidal, biomass, geothermal, Applications, Advantages & Disadvantages of Renewable Energy.			
CO2	Describe the fundamentals of solar cells.		
M2.01	Explain current conduction in semiconductor	3	Understanding
M2.02	Explain formation of P-N junction of Semiconductor	3	Understanding
M2.03	Describe the materials for Opto-Electronic applications.	3	Understanding
M2.04	Explain the concept of solar cell and its main elements.	3	Understanding
	Series Test - I	1	

Contents: Solar cell fundamentals: Current conduction in semiconductor, Atomic structure of silicon, Energy band formation in semiconductor, P-Type and N-type material with silicon Formation of P-N junction of Semiconductor, Principles for Electron-Hole Pair generation by Photon absorption, Photo- electric effect, Photo-conductive effect and Photovoltaic effect. Materials for Opto-Electronic applications. Concept of solar cell, Main elements of silicon solar cell.			
CO3	Explain the characteristics of photovoltaic cells and factors affecting solar cell performance.		
M3.01	Explain characteristics of Solar cells	4	Understanding
M3.02	Explain Design considerations of Solar cells	4	Understanding
M3.03	Explain Factors limiting the efficiency of solar cell.	4	Understanding
M3.04	Explain Impact of external parameters on solar cell performances	4	Understanding
Contents: Solar cell characteristics: Current-Voltage (I-V) characteristics of a Photovoltaic cell, Power-Voltage (P-V) characteristics of a Photovoltaic cell, Equivalent circuit of a solar cell, Maximum power point (MPP). Design considerations of Solar cells – Short circuit current (Isc), Open circuit voltage (Voc), Fill factor (FF), Energy losses & factors for loss, Efficiency. Factors limiting the efficiency of solar cell. Impact of external parameters on solar cell performances – (i)Radiation, (ii)Temperature, (iii) Wind velocity.			
CO4	Classify various photovoltaic cell technologies and describe the fabrication and formation of PV modules		
M4.01	Classify solar cells	4	Understanding
M4.02	Explain Materials used for Photovoltaic Cells	4	Understanding
M4.03	Explain Technologies for Photovoltaic Cells Fabrication	5	Understanding

M4.04	Describe the fabrication and formation of PV modules	5	Understanding
	Series Test - II	1	
Contents: Technologies for Photovoltaic Cells Fabrication: Classification of solar cell, Cell size. Single crystalline silicon cell, Polycrystalline silicon cell. Thin film solar cell – Amorphous Silicon, Gallium Arsenide, Cadmium Telluride, Copper Indium Gallium Diselenide. Dye-sensitized Solar Cell (DSSC) technology, Organic solar cell technology, Quantum Dot Solar cell technology. Concept of PV module, PV panel, PV array and its formation. Silicon Group and non-Silicon Group, PV cell, PV module, PV panel and PV array fabrication.			

Text /Reference:

T/R	Book Title/Author
T1	Fundamentals of Renewable Energy Sources by N.S. Rathore (Author), Khobragade Chetan (Author), Asnani Bhawana (Author)
T2	Fundamentals and Applications of Renewable Energy Mehmet Kanoğlu Yunus A. Çengel John M. Cimbali
R1	Fundamentals of Renewable Energy Sources by G.N. Tiwari
R2	Fundamentals of Renewable Energy Processes By Aldo Vieira da Rosa

Online Resources:

Sl. No.	Website Link
1	https://archive.nptel.ac.in/courses/115/107/115107116/